

ADITHYA BHASKAR

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EDUCATION

2023-28	Ph.D. in Computer Science, Princeton University, USA <i>Advised by Prof. Danqi Chen</i> (Specialization : Natural Language Processing) Research interests : LLM post-training, LLM tool-use, Mechanistic Interpretability	4.00/4.00
2019-23	Bachelor of Technology in Computer Science and Engineering (Honors), IIT Bombay, India	9.67/10.00
2017-19	High School, Central Board of Secondary Education, India	97.2/100.0
2017	Senior Secondary School, Central Board of Secondary Education, India	98.0/100.0

PUBLICATIONS

2025	Unintentional Unalignment : Likelihood Displacement in Direct Preference Optimization, Noam Razin, Sadhika Malladi, Adithya Bhaskar , Danqi Chen, Sanjeev Arora, and Boris Hanin	ICLR 2025
2024	Finding Transformer Circuits With Edge Pruning, Adithya Bhaskar , Alexander Wettig, Dan Friedman, and Danqi Chen	NeurIPS 2024 (Spotlight)
2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models, Adithya Bhaskar , Dan Friedman, and Danqi Chen	ACL 2024 (Oral)
2023	Benchmarking and Improving Text-to-SQL Generation under Ambiguity, Adithya Bhaskar* , Tushar Tomar*, Ashutosh Sathe, and Sunita Sarawagi	EMNLP 2023
2023	Prompted Opinion Summarization with GPT-3.5, Adithya Bhaskar , Alexander R. Fabbri, and Greg Durrett	ACL 2023 (Findings)
2023	Performance Bounds for LASSO under Multiplicative Noise : Applications to Pooled RT-PCR Testing, Richeek Das, Aaron Jerry Ninan, Adithya Bhaskar , and Ajit Rajwade	Signal Processing, Vol. 214

PREPRINTS

2025	Language Models That Think, Chat Better, Adithya Bhaskar* , Xi Ye*, and Danqi Chen	arXiv preprint, arXiv:2509.20357
2025	Cache Me If You Can : How Many KV's Do You Need for Effective Long-Context LMs?, Adithya Bhaskar* , Alexander Wettig*, Tianyu Gao, Yihe Dong, and Danqi Chen	arXiv preprint, arXiv:2506.17121
2025	Extracting Rule-based Descriptions of Attention Features in Transformers, Dan Friedman, Adithya Bhaskar , Alexander Wettig, and Danqi Chen	arXiv preprint, arXiv:2510.18148
2024	Continual Memorization of Factoids in Language Models, Howard Chen, Jiayi Geng, Adithya Bhaskar , Dan Friedman, and Danqi Chen	arXiv preprint, arXiv:2411.07175
2024	Improving Language Understanding from Screenshots, Tianyu Gao, Zirui Wang, Adithya Bhaskar , and Danqi Chen	arXiv preprint arXiv:2402.14073

INVITED TALKS

April 2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models <i>Host : Mathew Monfort</i>	Amazon AWS
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SCHOLASTIC ACHIEVEMENTS

2024	Recipient of the Hisashi and Masae Kobayashi *67 Fellowship .
2023	Recipient of the Thomas Dooie Class of 1974 Research Award .
2019	All India Rank 18 in JEE Advanced 2019 among 240 thousand candidates.
2019	All India Rank 114 in JEE Mains 2019 among 1.1 million candidates.
2018	Secured a position in the top 39 ranks in the Indian National Physics Olympiad and was invited to the Orientation-cum-Selection-Camp in Physics held in May-June 2018.
2018	Secured a position in the top 49 ranks in the Indian National Chemistry Olympiad and was invited to the Orientation-cum-Selection-Camp in Chemistry held in May-June 2018.
2016	Among the 39 students to clear the Indian National Mathematical Olympiad , becoming one of the youngest to ever be invited to the Orientation-cum-Selection-Camp in Mathematics aged 14 .

EXPERIENCE

UT Austin
Summer 2022

Research Intern, NATURAL LANGUAGE PROCESSING, USA

Advisor : Prof. Greg Durrett

Very Large Language Models for Multi-Document Summarization

- › Developed metrics to measure **factuality**, **faithfulness** and **specificity** (whether it is correct, prefers major viewpoints, and is not too generic) for a summary of multi-document text such as hotel reviews.
- › Utilized the above along with an n -gram abstractiveness metric to benchmark **GPT-3.5**, and showed that simple hierarchical summarization of large text performs poorly on faithfulness and specificity.
- › Investigated various **pre-clustering** and **pre-summarization** methods and illustrated that pre-summarization with a pretrained keyword-based extractive model improves correctness, faithfulness and specificity, while only marginally affecting abstractiveness.

Uppsala University
Summer 2021

Research Intern, FORMAL VERIFICATION,

Advisor : Prof. Parosh Abdulla

Model Checking for Programs Running under the ARMv8 Memory Model

- › Developed a **model** and **simulator** for programs running under the ARMv8 memory model, and demonstrated the equivalence of the model to the ARM specification using **7500+** litmus tests.
- › Used **Context Bounded Model Checking** to perform **State Reachability Analysis** for programs under the ARMv8 memory model, achieving **up to an order of magnitude** of speedup over existing checkers.

OTHER PROJECTS

Robust Models
Spring 2023

Bachelor's Project, NATURAL LANGUAGE PROCESSING, Guide : Prof. Sunita Sarawagi

Automated data augmentation for robustness.

- › Demonstrated that training a Text-to-SQL model on partially masked (underspecified) inputs leads to **diversity** in the model outputs, including in **columns/tables**, **string literals**, **integers**, and **aggregates**.
- › Filtered the outputs by model probabilities relative to the output with the unmasked question.
- › Furnished questions for the generated queries via an SQL-to-Text model. Data augmentation with the pairs led to increases in accuracy on the SPIDER dataset and the robustness benchmark, Dr. SPIDER.
- › The increases exceeded those obtained by augmenting with Dr. SPIDER style perturbations.

Group Testing
Fall 2022

R&D Project, COMPRESSIVE SENSING, Guide : Prof. Ajit Rajwade

Applying compressive sensing to improve COVID-19 Group Testing.

- › Proved the theoretical applicability of Compressive Sensing with Weighted LASSO for **any general noise model** with an asymptotically well-defined Moment-Generating-Function.
- › Derived values of the optimal weights for the case of **Multiplicative Gaussian Noise** as in RT-PCR tests.
- › Demonstrated improvements in **sensitivity**, **specificity**, **MCC** and **RMSE** by Monte Carlo Simulations.

C Decompiler
Fall 2020

Course Project, SOFTWARE SYSTEMS, Guide : Prof. Amitabha Sanyal

Recovering Code From Compiled RTL

- › Built a **decompiler** to convert **Register Transfer Language** to **C** for portability across architectures.
- › Utilized **lex** and **bison** to parse source code in RTL and identify program elements like **assignments**, **basic arithmetic operations**, **conditional/looping constructs**, **function calls** and **memory accesses**.
- › Performed **local & global data flow analysis** and **control flow analysis** to contextualize parsed code.

TEACHING

Spring 2025

Graduate Teaching Assistant, *COS 484 : Natural Language Processing*

Princeton University

Fall 2024

Graduate Teaching Assistant, *COS 597R : Deep Dive into Large Language Models*

Princeton University

Instructors : Danqi Chen and Sanjeev Arora

SERVICE

2025

NeurIPS 2025

Reviewer

2025

ICML 2025 MOSS Workshop

Reviewer

2024

NeurIPS 2024 ATTRIB Workshop

Reviewer